Nathan Xia

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Engineering Portfolio: Nathan Xia

Education

- **University of Toronto** — Toronto, Canada
  Bachelor of Applied Science - Engineering Science | Lachlan Dales McKeller Scholar — *September 2025 - Present*
  cGPA: 3.95/4.00

Skills Summary

- **Languages:** Python, Java, MATLAB, C, C++ (Arduino)
- **Tools:** ANSYS Fluent, Autodesk CFD, SolidWorks, Fusion 360, Autodesk Inventor, OpenRocket, OpenVSP
- **Skills:** CFD simulation and meshing, aerodynamic analysis and design, CAD, wind tunnel analysis and design, signal processing and analysis

Experience

- **U of T Institute for Aerospace Studies** — FCET Research Group
  Research Intern — *May 2026 - September 2026*
  - Selected as **1 of only 2 first-year students** to conduct research at UTIAS, independently leading an aeroacoustics study on remote microphone probe calibration for wind-tunnel pressure measurements. Found here: Preprint
  - Owned the full experimental workflow from apparatus and test-matrix design through **data acquisition and FFT-based signal processing** across 16 investigated variables, producing a **70+ GB dataset across 1,200+ collected data files.**
  - Developed a **MATLAB** acquisition and analysis pipeline and identified improvements that **reduced transfer function uncertainty by 69%** and **extended the usable frequency range to 1.5**× its original range.
  - Experimentally validated findings through **20+ trials** that informed the group's ongoing RMP research and future RMP projects, including improved calibration configurations and procedures.
- **Robotics for Space Exploration - Aerial** — University of Toronto
  Mechanical Co-Lead — *July 2026 - Present*
  - Direct all mechanical design and integration activities for the subsystem, with responsibility for architecture, **CAD**, analysis, manufacturing, assembly, and final system integration.
  - Lead a team of mechanical recruits, overseeing **CAD onboarding, CFD analysis,** design iteration, and manufacturing while reviewing and coordinating technical work across the team.

  Mechanical Sub-team Recruit — *September 2025 - July 2026*
  - Conceived the glider's lifting-body configuration, replacing a conventional fuselage with an airfoil-profile body; evaluated **20+ airfoils in OpenVSP/VSPAERO** to select the final geometry.
  - Integrated passive flow-control devices, including vortex generators and end plates, to mitigate boundary-layer separation, increasing simulated lift by **5× at 20 m/s.**
  - Designed and simulated rocket nose-cone geometries in Autodesk CFD, iterating through **30+ profiles** using CFD and rocket-performance simulations.
- **STEM Fellowship REO** — University of Calgary
  Student Lab Intern — *March 2025*
  - Selected as **1 of 30** from a competitive pool of ≈ 150 **applicants** for a research internship, collaborating with graduate researchers and professors on carbon-cycle and seawater CO_2 removal.
  - Performed chemical sample analysis using ion chromatography (IC), optical emission spectroscopy (OES), and scanning electron microscopy (SEM), and presented findings to ≈ **60 researchers** at the University of Calgary.

Projects

- **Lattice Boltzmann CFD Solver**: This project is ongoing. View Results.
  Developed and verified a Python-based D2Q9 Lattice Boltzmann CFD solver from scratch featuring configurable geometries and flow conditions, velocity, pressure, and vorticity visualization, and aerodynamic force tracking. Validated the solver using flow over a cylinder, **reproducing von Kármán vortex shedding** and the expected oscillations in lift and drag forces. Further benchmarked the solver on a NACA 0012 at $Re = 2000$ and $AOA = 8°$, **producing a Strouhal number within 1.7% of a published ANSYS Fluent result.**
- **Vortex Generator Research Project: Paper**
  Designed and conducted a research project optimizing vortex generators (VGs) for boundary layer separation mitigation on a NACA 2412 airfoil. Identified an optimal VG gap width and chordwise placement using a flow visualization wind tunnel, which was independently designed and manufactured. Optimal placement **increased the stall angle by 76%**, delaying the observed boundary layer separation significantly. The airfoil was mathematically designed on Desmos before being imported into CAD software to be edited. Presented findings at CYSF and CWSF. **1 of 15** projects selected to compete in CWSF out of ≈ 600 **total projects.** Bronze medal recipient at CWSF out of ≈ 350 projects.
- **Desktop Wind Tunnel**:
  Designed and fabricated a compact desktop wind tunnel for aerodynamic flow visualization, integrating a 3D-printed test section and diffuser, adjustable suction fan, and custom smoke-injection system. Iteratively refined the smoke-delivery geometry and airflow controls to improve tracer delivery and visibility within the test section.